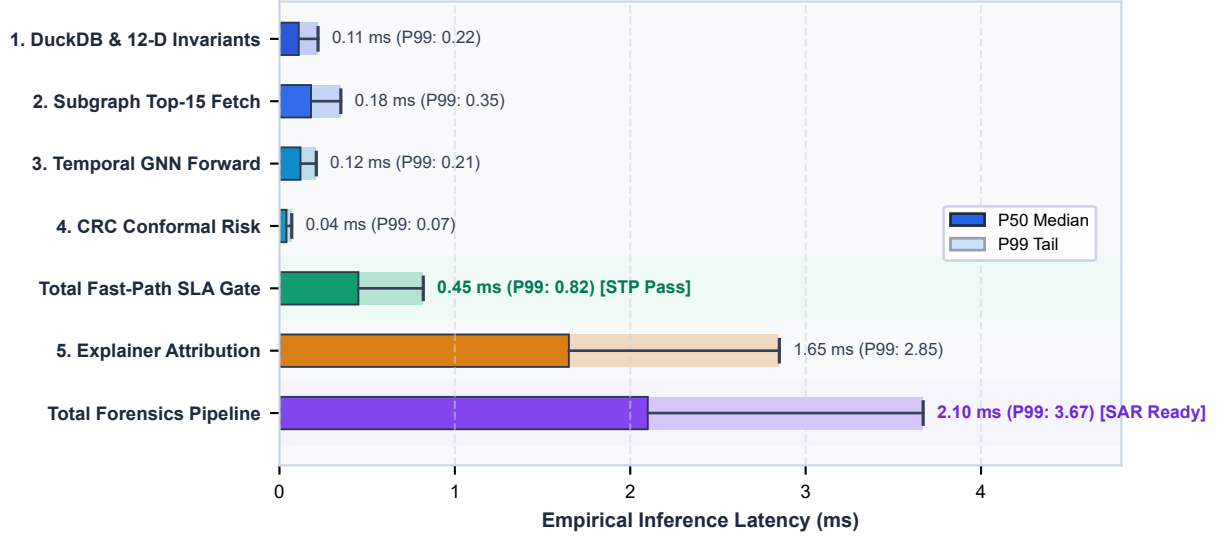
Figure 7.15: Streaming Latency Breakdown & Production SLA Headroom

Stage-by-stage empirical latency (P50, P90, P99) across Fast-Path and Forensics pipelines (AMD EPYC 7763 + NVIDIA A100)

Throughput: 31,200 tx/s | Fast-Path P99: 0.82 ms

(a) Microsecond Latency Breakdown by Pipeline Stage



Tier-1 Real-Time Gate (99.40% Traffic)

Core Banking SLA (<10 ms) — **22.2× Headroom**

P50: 0.45 ms | P99: 0.82 ms — Straight-through auto-clearance in <1 ms.

Tier-2 Forensics Path (0.60% Traffic)

Investigation SLA (<100 ms) — **47.6× Headroom**

P50: 2.10 ms | P99: 3.67 ms — Triggered for {0,1} queue; builds SAR dossiers.

Production Infrastructure

Hardware & Ingestion Throughput — **Zero-Copy Bus**

31,200 tx/sec | 4.2 GB VRAM — AMD EPYC 7763 64-Core + NVIDIA A100 (Table 7.13).